> Source data: 54 CARLA 0.9.16 scenes in Town10HD, 448 co-visible objects.
> Detectors: Ultralytics YOLO11m and RT-DETR-L evaluated over identical 108 frames via `python -m spike_dependence.contrast` . Decisions SD-022 through SD-024.

Contrast Condition: Evaluating Detector Architecture Homogeneity vs. Heterogeneity

To evaluate whether shared neural network weights explain inter-agent error correlation, a second detector of a fundamentally different architecture—the real-time vision transformer RT-DETR-L—was evaluated over the identical 108 rendered CARLA frames.

Because detection accuracy differs between YOLO11m and RT-DETR-L, comparison across homogeneous and heterogeneous configurations evaluates Pearson's ϕ correlation coefficient and odds ratios rather than the raw inflation ratio.

Condition	**Architecture Pairing**	ϕ **Correlation**	**Odds Ratio**	p_A	p_B
Homogeneous (YOLO11m / YOLO11m)	Dual CNN	0.374	5.129	0.368	0.348
Homogeneous (RT-DETR-L / RT-DETR-L)	Dual Transformer	0.379	6.605	0.234	0.214
Heterogeneous (YOLO11m / RT-DETR-L)	CNN + Transformer	0.278	3.889	0.368	0.214
Heterogeneous (RT-DETR-L / YOLO11m)	Transformer + CNN	0.292	4.003	0.234	0.348

Paired Bootstrap Analysis on Architecture Contrast

- **Mean Homogeneous Association (ϕ): 0.3766**
- **Mean Heterogeneous Association (ϕ): 0.2852**
- **Paired Association Difference ($\Delta\phi$): +0.0915**, 95% CI **[+0.0466, +0.1401]**

Empirical Findings

1. **Shared Model Weights Contribute to Co-Failure:** Homogeneous pairs exhibit higher error association than heterogeneous pairs. Because the confidence interval on the paired difference strictly excludes zero, deploying identical model weights measurably increases joint failure rates.
2. **Model Diversity Fails to Restore Independence:** Heterogeneous pairs running distinct architectures (CNN vs. vision transformer) remain strongly correlated ($\phi = 0.2852$, odds ratio ≈ 4.0). Shared weights are therefore a contributing channel rather than the primary driver of failure correlation; the shared physical embodiment dominates model-induced correlation.

This empirical finding broadens the research scope: the independence assumption fails even across heterogeneous agent teams, refuting the common assumption that architectural diversity alone guarantees uncorrelated perception errors.